

IISc Mech Eng 226 Applied Dynamics I

No calculators, books or notes allowed.

3 Problems, 3 hours

Final Exam

Thursday Dec 4, 2023.

9 AM to 12 PM.

*** How to get the highest score? ***

Please do these things:

- **Your written work.** Start each problem on a clean sheet.
— Put ***your name and *problem number** and ***page number** on the top of every side of every sheet.

✓ Draw **Free body diagrams** whenever force, moment, linear momentum, or angular momentum balance are used.

- Use correct **vector notation**.

A+ Be (I) neat, (II) clear and (III) well organized.

□ TIDILY REDUCE and box in your answers (Don't leave simplifiable algebraic expressions).

» Make appropriate Matlab code clear and correct.

You can use shortcut notation like " $\phi_7 = 2\pi$ " instead of, say, " $\text{phi}(7) = 2*\text{pi};$ ".

Small syntax errors will have small penalties.

↗ Clearly **define** any needed dimensions ($\ell, h, d, \dots$), coordinates ($x, y, r, \theta \dots$), variables ($v, m, t, \dots$), base vectors ($\hat{i}, \hat{j}, \hat{e}_r, \hat{e}_\theta, \hat{\lambda}, \hat{n} \dots$) and signs ($\pm$) with sketches, equations or words.

→| **Justify** your results so a grader can distinguish an informed answer from a guess. If you quote a fact that a grader might doubt your understanding of, explain it. Especially if it is not commonly used.

🐼 If a problem seems *poorly defined*, clearly state any reasonable assumptions (that do not oversimplify the problem).

≈ Work for **partial credit** (from 60–100%, depending on the problem)

- Put your answer in terms of well defined variables even if you have not substituted in the numerical values.
- Reduce the problem to a clearly defined set of equations to solve.
- Provide Matlab code which would generate the desired answer, and explain the nature of the output (unless specifically precluded).

Problem 1: ____/50

Problem 2: ____/50

Problem 3: ____/50

A short way of describing the exam questions below is the following: find the equations of motion for

- 1) a 2D rigid object with a skate constraint;
- 2) a point mass held back by a linear dashpot; and
- 3) a vertically shaken inverted pendulum.

1) Chaplygin sleigh. 2D. Gravity is not relevant. A sleigh is a single rigid object with mass m and moment of inertia I about its center of mass G. It moves freely on the xy plane except that there is a skate constraint at C. The skate is aligned with the line segment CG. The distance from C to G is d . At some instant you are given the position of point C, $(x_{C/0}, y_{C/0})$; the orientation of the line CG, θ measured counter-clockwise from the positive x axis; the speed of point C, $v_{C/0}$; and the angular velocity of the sleigh, $\bar{\omega} = \dot{\theta} \hat{k}$. Note, the velocity of point C is always in the direction of $\bar{r}_{CG} = \bar{r}_{G/C}$. Find $\ddot{\theta}$ in terms of some or all of $x_{C/0}, y_{C/0}, v_{C/0}, \theta, \dot{\theta}, I, m$ and d .

2) 2Dashpot and mass. 2D. No gravity. The only force on a point mass m is from a linear dashpot c connecting the mass to the origin. The dashpot tension is $T = c\dot{\ell}$ where c is a constant, T is the tension in the dashpot and ℓ is the distance from the origin to the mass.

- a) Which, if any, of these quantities is constant: Linear momentum, Angular momentum about 0, Total Energy, Kinetic Energy. As for all questions on this exam, explain your answers.
- b) In terms of some or all of $x, y, \dot{x}, \dot{y}, c$ and m , find $\ddot{x}$.

3) Shaken inverted pendulum. 2D. There is gravity g . One end of a stick with length ℓ is connected to an ideal hinge at moving point $0'$. A motor drives point $0'$ to move up and down according to this formula $\bar{r}_{0'/0} = \delta \sin(\omega t) \hat{j}$, where δ is the constant amplitude of shaking at constant frequency ω in the vertical $\hat{j}$ direction. The center of mass G of the stick is a distance d from $0'$. The mass of the stick is m and its moment of inertia about G is I . The angle θ is the angle of the stick, measured CCW from an upwards vertical line. In terms of some or all of $\theta, \dot{\theta}, m, d, \ell, I$ and g , find $\ddot{\theta}$.